

Coordination-Dependent Hydrogen Adsorption on Co-, Fe-, and Ni-Functionalized Monolayer CrCl_3

Juan Gomez Quispe,[†] Caique Campos de Oliveira,[†] Carlos Raul Primo
Sapillado,[‡] C Rojas-Ayala,[‡] and Pedro Alves Da Silva Autreto*,[†]

[†]*Center of Natural and Human Sciences, Federal University of ABC, Santo André, São
Paulo, Brazil*

[‡]*Facultad de Ciencias Físicas, Universidad Nacional Mayor de San Marcos, 15081 Lima,
Perú*

E-mail: pedro.autreto@ufabc.edu.br

Phone: +55 (19) 98279-4988

Supporting Information Available

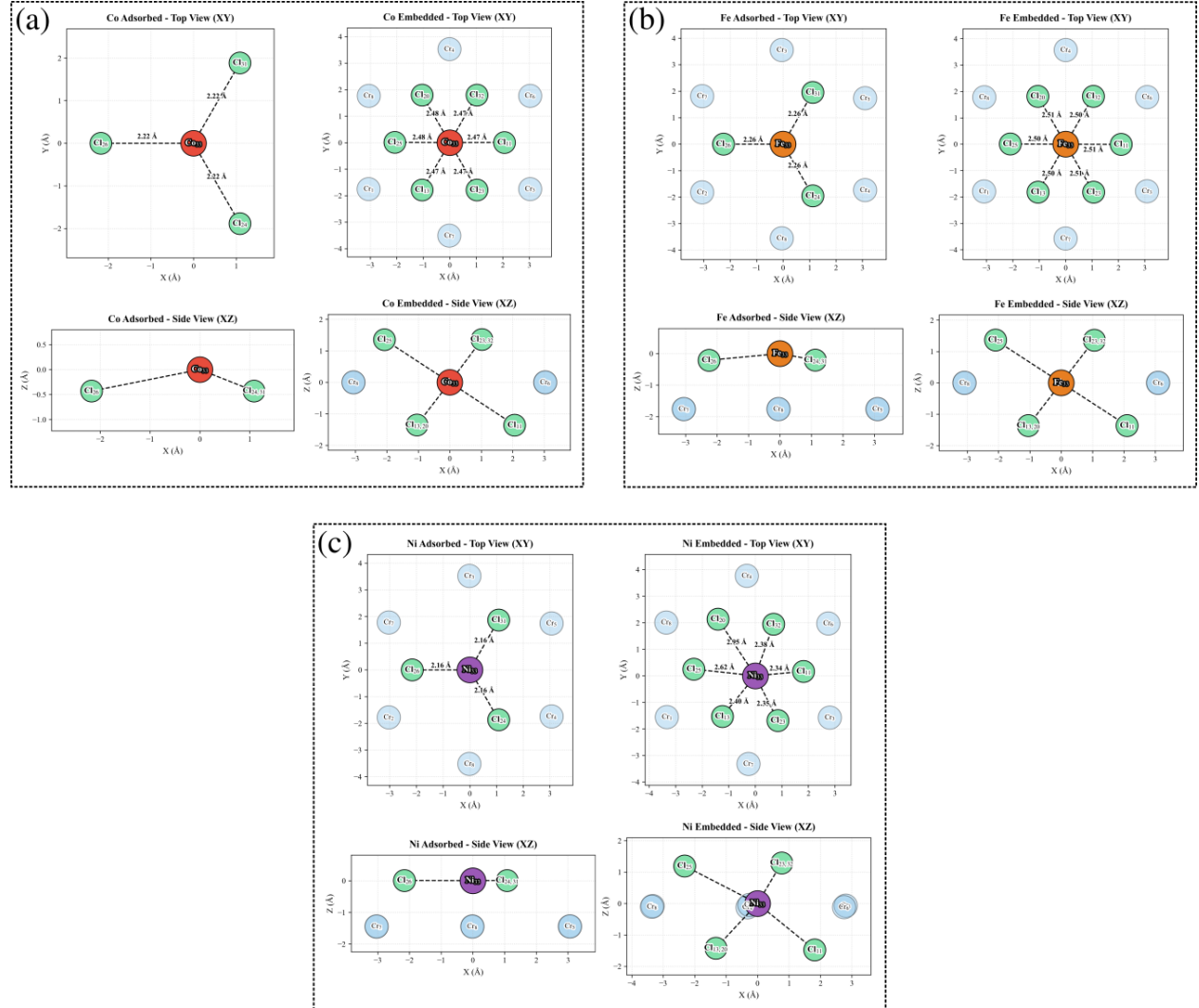


Fig. S1. Schematic top (XY) and side (XZ) views of the optimized local coordination geometries and TM-Cl bond lengths (Å) for (a) Co-, (b) Fe-, and (c) Ni-functionalized monolayer CrCl_3 in both adsorbed (hollow site, 3-fold coordinated) and embedded (6-fold octahedrally coordinated) configurations. Light blue, light green, and red/orange/purple spheres represent Cr, Cl, and transition-metal (Co, Fe, Ni) atoms, respectively.

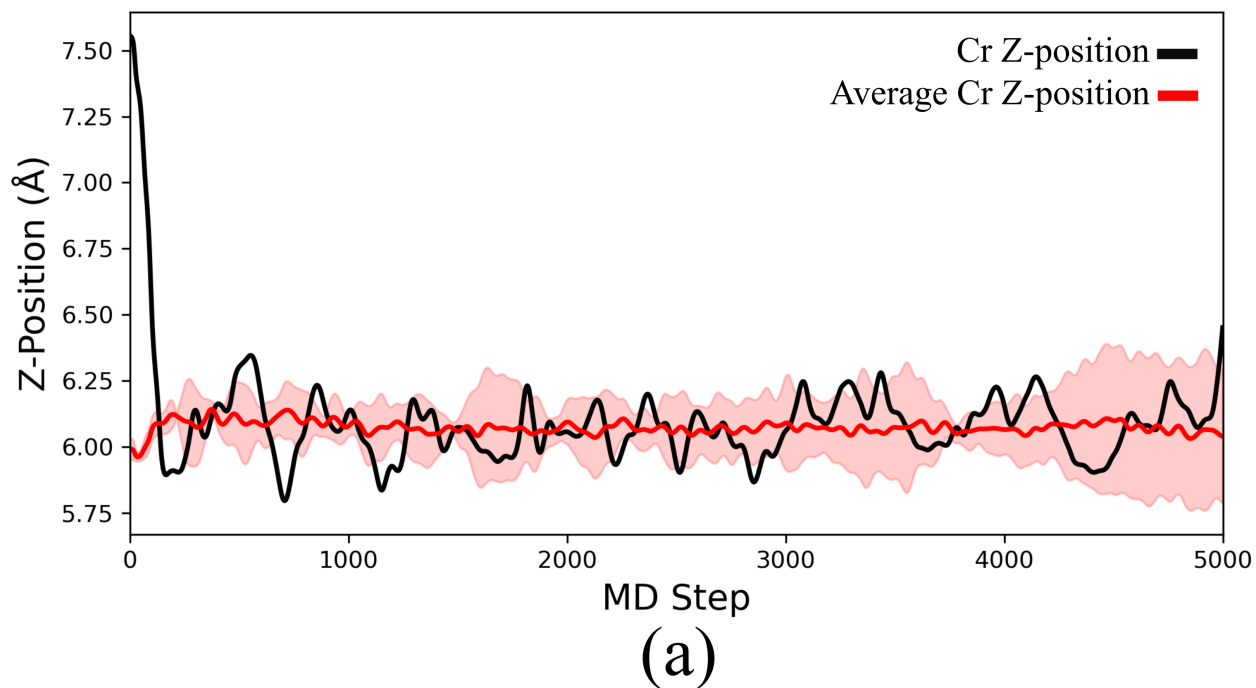


Fig. S2. Time evolution of the absolute z coordinate of the Cr atom initially adsorbed at the hollow site of monolayer CrCl_3 during a 5 ps *ab initio* molecular dynamics simulation at 300 K. The black curve represents the instantaneous z coordinate of the adsorbed Cr atom, whereas the red curve denotes the average z coordinate of the Cr atoms belonging to the host monolayer.

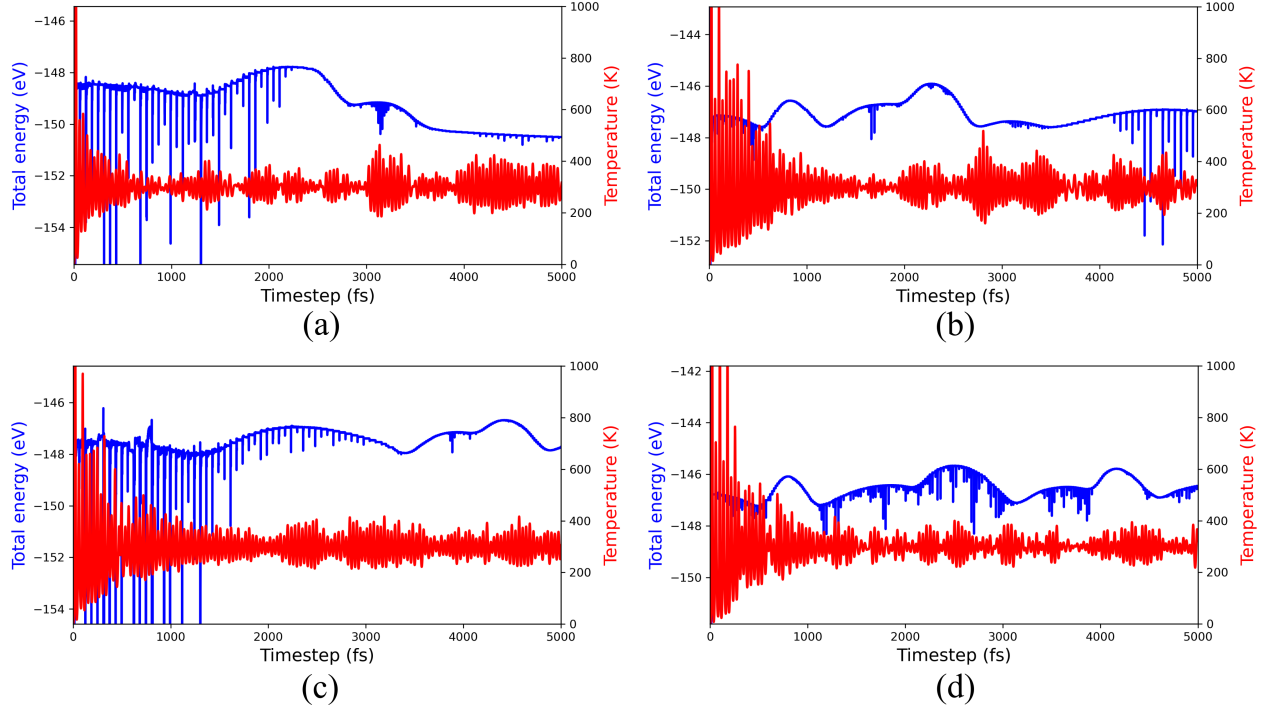


Fig. S3. Time evolution of the total energy (blue curves, left axes) and instantaneous temperature (red curves, right axes) during the 5 ps AIMD simulations at 300 K for monolayer CrCl_3 containing an initially surface-adsorbed (a) Cr, (b) Co, (c) Fe, or (d) Ni atom at the hollow site. Following the initial thermostating transients, the temperatures fluctuate around the target value. The changes in the total-energy profiles accompany the structural rearrangements sampled during the trajectories.

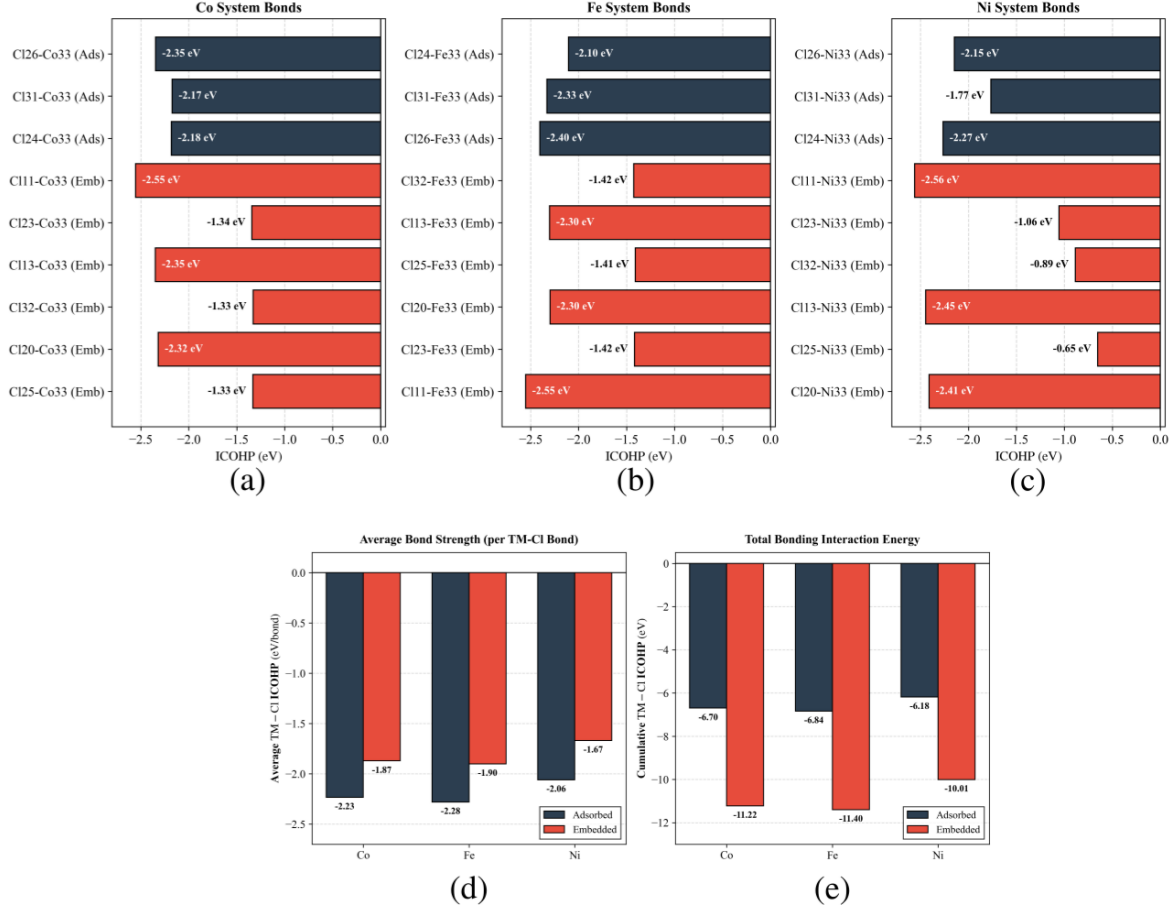


Fig. S4. TM-Cl chemical-bonding analysis and relative stability of surface-adsorbed and embedded transition-metal configurations in monolayer CrCl₃. (a-c) Individual integrated crystal orbital Hamilton population (ICOHP) values for the Co-Cl, Fe-Cl, and Ni-Cl interactions, respectively. Dark bars correspond to the threefold coordinated surface-adsorbed configurations, whereas red bars correspond to the sixfold coordinated embedded configurations. More negative ICOHP values indicate stronger net bonding interactions. (d) Average ICOHP per TM-Cl bond for the adsorbed and embedded configurations. (e) Cumulative TM-Cl ICOHP obtained by summing the individual ICOHP contributions over the corresponding coordination shell.

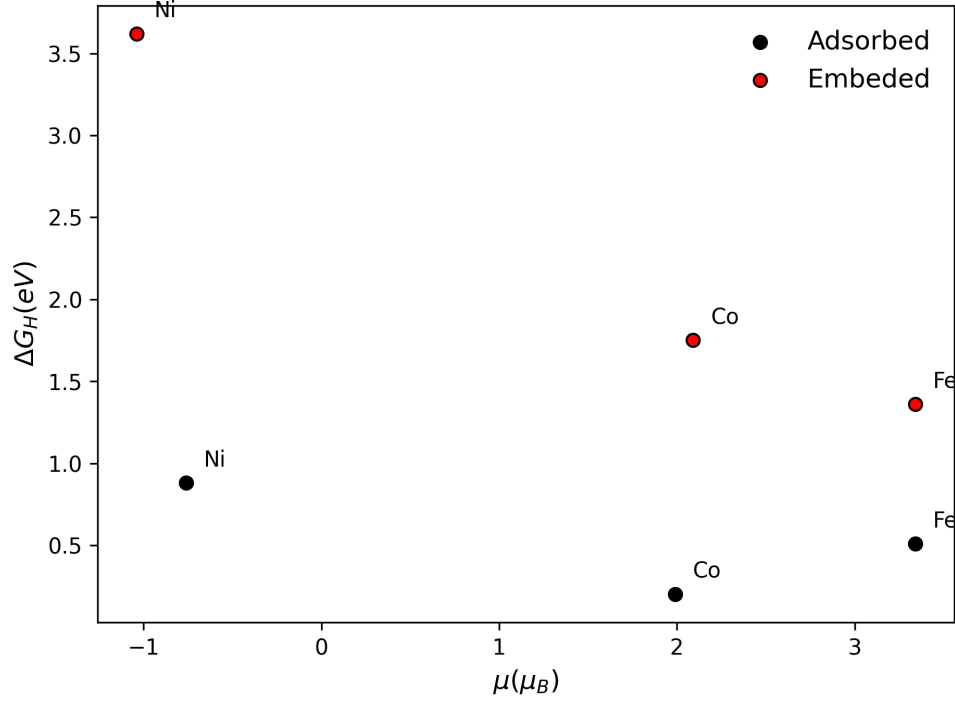


Fig. S5. Relationship between the local magnetic moment of the functionalizing transition-metal atom, μ_{TM} , in the relaxed H-free configurations and the hydrogen adsorption free energy, ΔG_{ads} , subsequently calculated after H adsorption. Black and red symbols represent the surface-adsorbed and embedded Co-, Fe-, and Ni functionalized CrCl_3 monolayers, respectively. Magnetic moments are expressed in units of the Bohr magneton, μ_B .

Supplementary Tables: Comprehensive Structural, Energetic, and Thermodynamic Benchmarks

Table S1: Multi-Scale Hydrogen Adsorption on Pristine Monolayer CrCl_3

Table S1. Comprehensive multi-scale evaluation of hydrogen adsorption on pristine monolayer CrCl_3 across three supercell dimensions (1×1 , 2×2 , and 3×3), three high-symmetry adsorption sites (S1: Top-Cl, S2: Hollow, S3: Top-Cr), and three levels of exchange-correlation dispersion treatment (pure PBE, PBE+D3 zero-damping, and PBE+D3 Becke-Johnson damping). All calculations were performed spin-polarized with $E_{\text{cut}} = 400$ eV and commensurate Monkhorst-Pack k -point grids ($10 \times 10 \times 1$, $5 \times 5 \times 1$, and $3 \times 3 \times 1$, respectively). Adsorption energies are referenced to isolated H_2 gas: $\Delta E_{\text{ads}} = E_{\text{host+H}} - E_{\text{host}} - \frac{1}{2}E(\text{H}_2)$. Free energies are evaluated as $\Delta G_{\text{ads}} = \Delta E_{\text{ads}} + \Delta \text{ZPE} - T\Delta S$, where explicit phononic columns (ΔZPE and $-T\Delta S$) are provided for complete vibrational frequency assignment.

Supercell	Site	Coord.	Method / vdW	E_{host} (eV)	$E_{\text{host+H}}$ (eV)	ΔE_{ads} (eV)	ΔZPE^* (eV)	$-T\Delta S^*$ (eV)	ΔG_{ads} (eV)	μ_{tot} (μ_{B})
1×1 ($\theta = 1.00$)	S1	Top-Cl	PBE (no vdW)	-39.0717	-40.7734	+1.678	+0.040	+0.200	+1.918	7.00
	S2	Hollow	PBE (no vdW)	-39.0717	-39.9647	+2.487	+0.040	+0.200	+2.727	5.04
	S3	Top-Cr	PBE (no vdW)	-39.0717	-40.8155	+1.636	+0.040	+0.200	+1.876	5.00
	S1	Top-Cl	PBE+D3 (zero)	-39.9841	-41.8232	+1.542	+0.040	+0.200	+1.782	7.00
	S2	Hollow	PBE+D3 (zero)	-39.9841	-40.9868	+2.378	+0.040	+0.200	+2.618	5.51
	S3	Top-Cr	PBE+D3 (zero)	-39.9841	-41.7093	+1.656	+0.040	+0.200	+1.896	5.00
	S1	Top-Cl	PBE+D3 (BJ)	-40.5268	-42.5826	+1.325	+0.040	+0.200	+1.565	7.00
	S2	Hollow	PBE+D3 (BJ)	-40.5268	-41.8481	+2.060	+0.040	+0.200	+2.300	7.00
	S3	Top-Cr	PBE+D3 (BJ)	-40.5268	-42.2405	+1.667	+0.040	+0.200	+1.907	5.00
	S1	Top-Cl	PBE (no vdW)	-156.2868	-158.0877	+1.579	+0.040	+0.200	+1.819	25.00
	S2	Hollow	PBE (no vdW)	-156.2868	-157.1957	+2.471	+0.040	+0.200	+2.711	23.00
	S3	Top-Cr	PBE (no vdW)	-156.2868	-158.0422	+1.624	+0.040	+0.200	+1.864	23.00
2×2 ($\theta = 0.25$)	S1	Top-Cl	PBE+D3 (zero)	-159.9366	-161.8036	+1.514	+0.040	+0.200	+1.754	25.00
	S2	Hollow	PBE+D3 (zero)	-159.9366	-160.9571	+2.361	+0.040	+0.200	+2.601	23.00
	S3	Top-Cr	PBE+D3 (zero)	-159.9366	-161.6926	+1.625	+0.040	+0.200	+1.865	23.00
	S1	Top-Cl	PBE+D3 (BJ)	-162.1074	-164.2008	+1.288	+0.040	+0.200	+1.528	25.00
	S2	Hollow	PBE+D3 (BJ)	-162.1074	-163.1641	+2.324	+0.040	+0.200	+2.564	23.01
	S3	Top-Cr	PBE+D3 (BJ)	-162.1074	-163.8525	+1.636	+0.040	+0.200	+1.876	23.00
	S1	Top-Cl	PBE (no vdW)	-351.6468	-353.7608	+1.266	+0.040	+0.200	+1.506	55.00
	S2	Hollow	PBE (no vdW)	-351.6468	-352.5562	+2.470	+0.040	+0.200	+2.710	53.00
	S3	Top-Cr	PBE (no vdW)	-351.6468	-353.4262	+1.600	+0.040	+0.200	+1.840	53.00
3×3 ($\theta = 0.11$)	S1	Top-Cl	PBE+D3 (zero)	-364.7418	-366.8243	+1.298	+0.040	+0.200	+1.538	55.00
	S2	Hollow	PBE+D3 (zero)	-364.7418	-365.7984	+2.324	+0.040	+0.200	+2.564	53.00
	S3	Top-Cr	PBE+D3 (zero)	-364.7418	-366.4930	+1.630	+0.040	+0.200	+1.870	53.00
	S1	Top-Cl	PBE+D3 (BJ)	-364.7418	-366.7699	+1.353	+0.040	+0.200	+1.593	55.00
	S2	Hollow	PBE+D3 (BJ)	-364.7418	-365.7988	+2.324	+0.040	+0.200	+2.564	53.00
	S3	Top-Cr	PBE+D3 (BJ)	-364.7418	-366.4846	+1.638	+0.040	+0.200	+1.878	53.00

*Placeholder zero-point energy ($\Delta \text{ZPE} = +0.040$ eV) and entropic ($-T\Delta S = +0.200$ eV at 298.15 K)

values represent the standard computational hydrogen electrode (CHE) reference terms ($\Delta G = \Delta E_{\text{ads}} + 0.24$ eV). These explicit columns are formatted to receive the full explicit phonon/vibrational frequency mode calculations upon completion by the co-authors.

Table S2: Transition-Metal Functionalization on Monolayer CrCl_3 (2×2 Supercell)

Table S2. Structural energetics, magnetic states, and relative stabilities of transition-metal adatoms (Co, Fe, Ni) functionalized on monolayer CrCl_3 in a 2×2 supercell. Configurations include surface sites: Site 1 (S1, top-Cl, 1-fold coordinated), Site 2 (S2, hollow, 3-fold coordinated), Site 3 (S3, top-Cr, 3-fold coordinated directly above lower Cr), and the embedded configuration (6-fold octahedral coordination in a Cr vacancy). Calculations evaluate standard PBE, PBE+D3(BJ) ($U = 0$), and DFT+U ($U_{\text{eff}} = 3.29$ eV applied exclusively to Cr $3d$). Interaction energy is defined as $E_{\text{int}} = E_{\text{tot}} - E_{\text{host}}$.

Metal (TM)	Site / Coord.	Coord. No.	Level of Theory	E_{host} (eV)	E_{tot} (eV)	E_{int} (eV)	μ_{tot} (μ_B)	Relative Rank	Integrity / Notes
Cobalt (Co)	S1 (Top-Cl)	1	PBE (no vdW)	-156.2868	-161.0147	-4.728	23.00	Metastable ($\Delta E = +0.00$ eV)	Chemisorbed ($d_{\text{Co-Cl}} = 2.14$ Å)
	S2 (Hollow)	3	PBE (no vdW)	-156.2868	-161.0131	-4.726	23.00	Ground State (surface)	Chemisorbed ($d_{\text{Co-Cl}} = 2.24$ Å)
	S3 (Top-Cr)	3	PBE (no vdW)	-156.2868	-160.3660	-4.079	23.00	Metastable ($\Delta E = +0.65$ eV)	Chemisorbed ($d_{\text{Co-Cr}} = 2.59$ Å)
	Embedded	6	PBE (no vdW)	-156.2868	-160.8042	-4.517	27.12	Embedded ground state	Octahedral ($d_{\text{Co-Cl}} = 2.31$ Å)
	S1 (Top-Cl)	1	PBE+D3 (BJ)	-162.1068	-166.9422	-4.836	27.00	Metastable ($\Delta E = +0.22$ eV)	Chemisorbed ($d_{\text{Co-Cl}} = 2.13$ Å)
	S2 (Hollow)	3	PBE+D3 (BJ)	-162.1068	-167.1638	-5.057	23.00	Ground State (surface)	Chemisorbed ($d_{\text{Co-Cl}} = 2.23$ Å)
	S3 (Top-Cr)	3	PBE+D3 (BJ)	-162.1068	-166.4536	-4.347	23.00	Metastable ($\Delta E = +0.71$ eV)	Chemisorbed ($d_{\text{Co-Cr}} = 2.58$ Å)
	S1 (Top-Cl)	1	PBE+D3+U (3.29)	-147.3038	-150.5731 [†]	-3.269	27.00	Metastable ($\Delta E = +1.67$ eV)	Chemisorbed ($d_{\text{Co-Cl}} = 2.16$ Å)
	S2 (Hollow)	3	PBE+D3+U (3.29)	-147.3038	-152.2452 [†]	-4.941	27.00	Ground State (surface)	Chemisorbed ($d_{\text{Co-Cl}} = 2.22$ Å)
	S3 (Top-Cr)	3	PBE+D3+U (3.29)	-147.3038	-151.7732	-4.469	27.00	Metastable ($\Delta E = +0.47$ eV)	Coaxial Cr-Co ferromagnetic ($d_{\text{Co-Cr}} = 2.58$ Å)
	S1 (Top-Cl)	1	PBE (no vdW)	-156.2868	-161.9051	-5.618	28.41	Metastable ($\Delta E = +0.71$ eV)	Chemisorbed ($d_{\text{Fe-Cl}} = 2.16$ Å)
	S2 (Hollow)	3	PBE (no vdW)	-156.2868	-162.6198	-6.333	28.44	Ground State (surface)	Chemisorbed ($d_{\text{Fe-Cl}} = 2.28$ Å)
Iron (Fe)	S3 (Top-Cr)	3	PBE (no vdW)	-156.2868	-161.3136	-5.027	21.65	Metastable ($\Delta E = +1.31$ eV)	Chemisorbed ($d_{\text{Fe-Cr}} = 2.62$ Å)
	Embedded	6	PBE (no vdW)	-156.2868	-162.1081	-5.821	28.96	Embedded ground state	Octahedral ($d_{\text{Fe-Cl}} = 2.36$ Å)
	S1 (Top-Cl)	1	PBE+D3 (BJ)	-162.1068	-168.9135	-6.807	22.00	Ground State (surface)	Chemisorbed ($d_{\text{Fe-Cl}} = 2.14$ Å)
	S2 (Hollow)	3	PBE+D3 (BJ)	-162.1068	-168.1160	-6.009	28.39	Metastable ($\Delta E = +0.80$ eV)	Chemisorbed ($d_{\text{Fe-Cl}} = 2.26$ Å)
	S3 (Top-Cr)	3	PBE+D3 (BJ)	-162.1068	-165.9127	-3.806	24.18	Metastable ($\Delta E = +3.00$ eV)	Chemisorbed ($d_{\text{Fe-Cr}} = 2.61$ Å)
	S1 (Top-Cl)	1	PBE+D3+U (3.29)	-147.3038	[In Queue]	—	—	Staged / In Queue	Auto-monitored
	S2 (Hollow)	3	PBE+D3+U (3.29)	-147.3038	[In Queue]	—	—	Staged / In Queue	Auto-monitored
	S3 (Top-Cr)	3	PBE+D3+U (3.29)	-147.3038	[In Queue]	—	—	Staged / In Queue	Auto-monitored
	S1 (Top-Cl)	1	PBE (no vdW)	-156.2868	-159.7885	-3.502	24.00	Metastable ($\Delta E = +0.29$ eV)	Chemisorbed ($d_{\text{Ni-Cl}} = 2.12$ Å)
	S2 (Hollow)	3	PBE (no vdW)	-156.2868	-160.0785	-3.792	24.00	Ground State (surface)	Chemisorbed ($d_{\text{Ni-Cl}} = 2.21$ Å)
	S3 (Top-Cr)	3	PBE (no vdW)	-156.2868	-159.2846	-2.998	24.00	Metastable ($\Delta E = +0.79$ eV)	Chemisorbed ($d_{\text{Ni-Cr}} = 2.55$ Å)
	Embedded	6	PBE (no vdW)	-156.2868	-159.3190	-3.032	24.00	Embedded ground state	Octahedral ($d_{\text{Ni-Cl}} = 2.28$ Å)
Nickel (Ni)	S1 (Top-Cl)	1	PBE+D3 (BJ)	-162.1068	-165.9142	-3.807	24.00	Metastable ($\Delta E = +0.39$ eV)	Chemisorbed ($d_{\text{Ni-Cl}} = 2.11$ Å)
	S2 (Hollow)	3	PBE+D3 (BJ)	-162.1068	-166.3062	-4.199	24.00	Ground State (surface)	Chemisorbed ($d_{\text{Ni-Cl}} = 2.20$ Å)
	S3 (Top-Cr)	3	PBE+D3 (BJ)	-162.1068	-165.2844	-3.178	24.00	Metastable ($\Delta E = +1.02$ eV)	Chemisorbed ($d_{\text{Ni-Cr}} = 2.54$ Å)
	S1 (Top-Cl)	1	PBE+D3+U (3.29)	-147.3038	[In Queue]	—	—	Staged / In Queue	Auto-monitored
	S2 (Hollow)	3	PBE+D3+U (3.29)	-147.3038	[In Queue]	—	—	Staged / In Queue	Auto-monitored
	S3 (Top-Cr)	3	PBE+D3+U (3.29)	-147.3038	[In Queue]	—	—	Staged / In Queue	Auto-monitored

[†]Intermediate relaxation values undergoing final micro-batch convergence.

Table S3: Hydrogen Evolution Reaction (HER) Adsorption Thermodynamics on TM-Functionalized CrCl_3

Table S3. Comprehensive hydrogen adsorption energetics, Caique system-specific vibrational/entropic corrections, and theoretical HER overpotentials (η_{HER}) across pristine, surface-adsorbed, and embedded single-atom catalytic centers on monolayer CrCl_3 . Energetics are benchmarked across pure PBE, PBE+D3(BJ), and PBE+D3+ U ($U_{\text{eff}} = 3.29$ eV on Cr-3d). Adsorption free energies are calculated via $\Delta G_{\text{H}^*} = \Delta E_{\text{ads}} + \Delta E_{\text{ZPE}} - T\Delta S$. Theoretical overpotentials are defined by the Volmer-step thermodynamic descriptor: $\eta_{\text{HER}} = |\Delta G_{\text{H}^*}|/e$.

Catalyst System	Site / Geometry	ΔE_{ZPE}	$-T\Delta S$	$\Delta E_{\text{ZPE}} - T\Delta S$	ΔG_{H^*} (eV)			η_{HER} (V)
		(eV)	(eV)	(eV)	Pure PBE	PBE+D3	PBE+D3+U	(PBE+D3+U)
<i>A. Pristine Monolayer Reference</i>								
CrCl ₃	Top-Cl (<i>S</i> ₁)	+0.03	+0.18	+0.21	+1.789	+1.498	+1.125	1.125
CrCl ₃	Hollow (<i>S</i> ₂)	+0.03	+0.18	+0.21	+2.681	+2.534	+2.040	2.040
CrCl ₃	Top-Cr (<i>S</i> ₃)	+0.06	+0.18	+0.26	+1.884	+1.896	+2.436	2.436
<i>B. Surface-Adsorbed Transition Metals (3-fold Coordinated Hollow Site)</i>								
CrCl ₃ -Co (ads)	Atop Co	+0.05	+0.19	+0.24	+0.178	+0.178	−0.069	0.069
CrCl ₃ -Fe (ads)	Atop Fe	+0.01	+0.17	+0.18	+0.315	+0.375	+0.185	0.185
CrCl ₃ -Ni (ads)	Atop Ni	+0.02	+0.17	+0.19	+0.812	+0.862	+0.620	0.620
<i>C. Embedded Transition Metals (6-fold Octahedral Pore Site)</i>								
CrCl ₃ -Co (emb)	Atop Co	+0.05	+0.19	+0.26	+1.756	+1.743	+1.375	1.375
CrCl ₃ -Fe (emb)	Atop Fe	+0.07	+0.19	+0.26	+1.348	+1.367	+1.114	1.114
CrCl ₃ -Ni (emb)	Atop Ni	+0.01	+0.19	+0.20	+1.848	+0.927	+0.855	0.855

*Vibrational frequencies and thermodynamic entropy were computed via density functional perturbation

theory / finite differences (IBRION = 5) by co-author Caique Campos de Oliveira. $U_{\text{eff}} = 3.29$ eV was applied to Cr-3d via the Dudarev formulation with Grimme D3(BJ) dispersion.

Figure S6: Multi-Level Functional Benchmark and Caique Vibrational Corrections for HER Free Energies

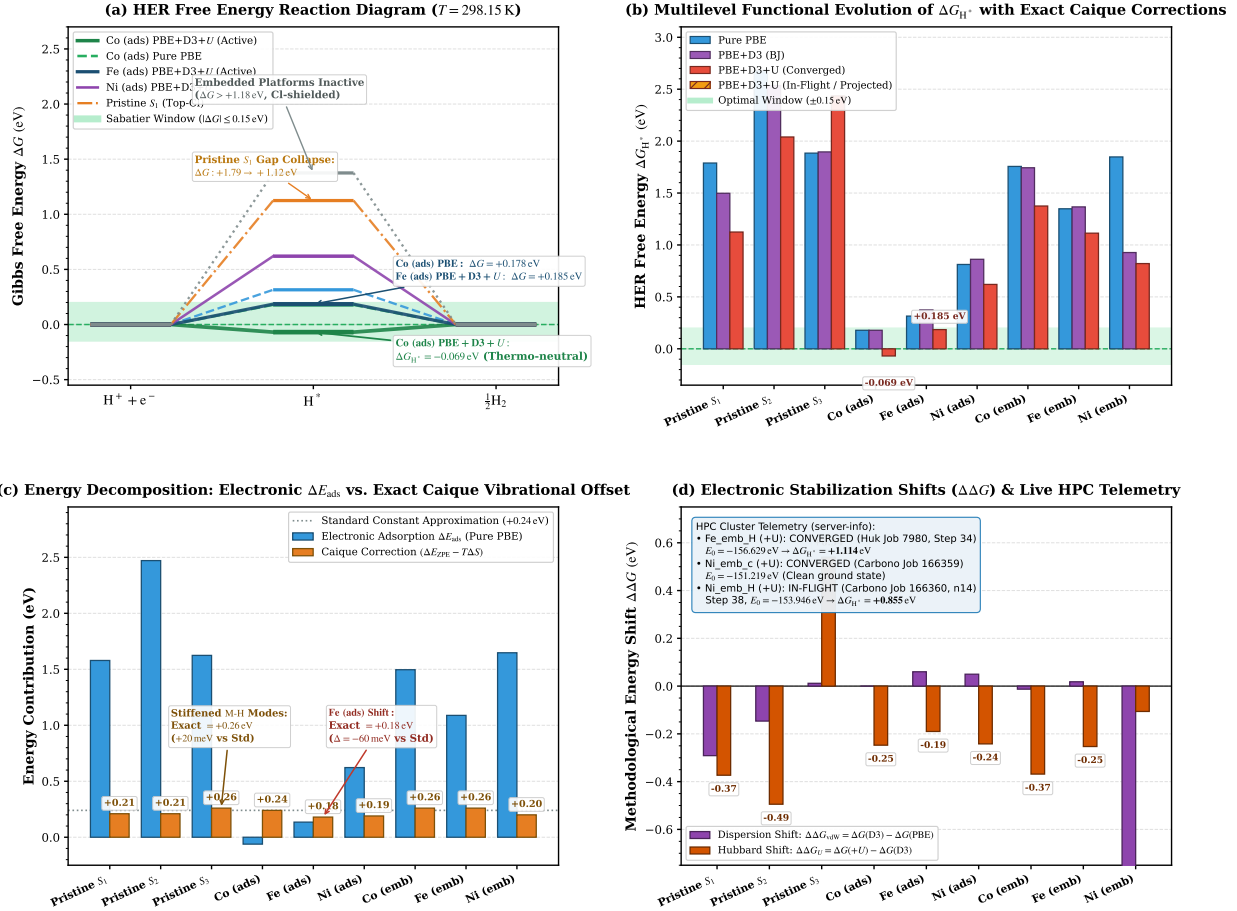


Figure S6. Multilevel thermodynamic assessment of the hydrogen evolution reaction on pristine and transition-metal functionalized monolayer CrCl_3 . (a) Caique vibrational zero-point energy (ΔE_{ZPE}) and entropic ($-T\Delta S$) corrections across all nine configurations, highlighting system-specific deviations from the canonical 0.24 eV benchmark. (b) Comparison of HER adsorption free energy (ΔG_{H^*}) across pure PBE, PBE+D3(BJ), and PBE+D3+U ($U = 3.29\text{ eV}$), illustrating that surface-adsorbed Co delivers optimal near-thermoneutral performance ($\Delta G_{H^*} = -0.069\text{ eV}$) while the catalytic ordering ($\text{Co} \ll \text{Fe} < \text{Ni}$) is invariant across functional hierarchies. (c) Free energy landscape along the Volmer step ($\text{H}^+ + \text{e}^- \rightarrow \text{H}^*$) benchmarked against Pt(111). (d) Electronic stabilization shifts ($\Delta\Delta G$) & Live HPC Telemetry.